

**A PROJECT REPORT
ON
BOOKSPHERE**

An Intelligent Book Discovery, Personal Library, and Community Platform

Submitted in partial fulfillment of the requirements for the award of the degree of
MASTER OF COMPUTER APPLICATIONS (MCA)

BY
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[ACADEMIC YEAR: 2025–2026]

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[HOD NAME AND DESIGNATION]
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Abstract

In the modern digital era, the proliferation of published literature has created an unprecedented paradox of choice for avid readers, researchers, and book enthusiasts. Traditional digital library catalogs and commercial bookstore interfaces frequently suffer from fragmented architectures, opaque algorithmic recommendations, disconnected community discussions, and bloated software stacks that compromise responsiveness and user privacy.

This project presents **BookSphere**, an intelligent, responsive, and secure web application designed to unify book discovery, personal reading library management, transparent recommendation generation, and literary community engagement within a cohesive, lightweight software architecture. Developed using pure object-oriented PHP (PHP 8.2+) and a clean Model-View-Controller (MVC) design pattern without heavy third-party framework overhead, BookSphere provides a high-performance, maintainable web system operating over a structured SQLite relational database.

At the core of BookSphere is **Recommendation Engine V2**, a deterministic, multi-source hybrid recommendation engine. Unlike opaque deep learning models or generic collaborative filters that suffer from cold-start latency and opacity, Recommendation Engine V2 synthesizes eight distinct signal sources (Sources A through H)—including direct category affinity, author affinity, wishlist saves, highly-rated reviews, recently viewed items, followed author releases, community engagement signals, and global popularity baselines. Candidates are scored using parameterized SQL expressions mirrored by PHP computation, filtered through strict library exclusion logic, and diversified using Maximal Marginal Relevance (MMR) reranking to mitigate genre over-concentration. Crucially, every recommendation produces a human-readable, deterministic explanation describing exactly why a title was selected.

BookSphere incorporates comprehensive personal reading management (tracking "Want to Read", "Currently Reading", and "Finished" states with custom ratings and reviews), an integrated community discussion hub with threaded replies, author following with notification dispatch, and administrative curation tools featuring Google Books API synchronization and content moderation. The application implements rigorous security controls, including synchronizer-token CSRF protection, secure session management,

role-based authorization middleware, and strict PDO prepared-statement SQL injection defenses.

Comprehensive testing and system verification—encompassing 64 automated CLI test suites, 12 Recommendation V2 suites, SQLite foreign key and integrity verifications, and headless Chrome/Edge DevTools Protocol (CDP) frontend audits—demonstrate that BookSphere delivers sub-160 millisecond response times, zero uncaught client-side JavaScript errors, and flawless responsive performance across desktop, tablet, and mobile viewports.

CHAPTER 1 — INTRODUCTION

1.1 Introduction

Reading remains one of humanity's most enduring intellectual pursuits. However, the contemporary digital landscape presents significant hurdles for readers attempting to navigate the vast universe of available literature. While millions of books have been digitized and cataloged, the software platforms built to facilitate discovery, personal organization, and community interaction remain fundamentally fragmented.

Readers commonly encounter two undesirable extremes:

1. **Commercial E-Commerce Platforms**: Websites designed primarily to drive transactional purchases rather than cultivate intellectual engagement. These systems employ opaque, profit-driven recommendation algorithms that promote sponsored bestsellers rather than matching readers' nuanced personal tastes, while cluttering interfaces with commercial advertisements.
2. **Outdated Social Cataloging Networks**: Legacy platforms burdened by decades of technical debt, sluggish user interfaces, bloated client-side JavaScript bundles, and disconnected social discussion boards that fail to integrate personal reading milestones with algorithmic discovery.

BookSphere was conceived and engineered to resolve this dichotomy. It is an intelligent, privacy-respecting, and highly responsive web application that harmonizes book discovery, catalog management, personalized recommendations, and community interaction within a unified, modern web architecture. Built on a clean, dependency-light PHP Model-View-Controller (MVC) foundation and backed by a robust SQLite relational database, BookSphere demonstrates that high-performance web systems do not require complex, resource-heavy microservices or opaque black-box machine learning models to deliver sophisticated personalization.

1.2 Problem Statement

Contemporary web platforms serving readers suffer from several critical architectural, functional, and user-experience limitations:

1. ****Opaque and Ineffective Recommendation Systems****: Existing book platforms predominantly deploy generic collaborative filtering or commercial black-box recommendation models. These systems frequently suffer from severe "cold-start" degradation for new users, create repetitive "filter bubbles" that continuously surface the same bestselling titles, and fail to explain **why** a particular book was recommended, eroding user trust.
2. ****Fragmented User Experience****: Readers are forced to use multiple disparate applications: one for discovering new titles, another for tracking their personal reading progress and library statuses ("Want to Read", "Currently Reading", "Finished"), a third for writing literary reviews, and external forums or social media for literary discussions.
3. ****Bloated Web Architectures and Sluggish Performance****: Many modern web applications depend on excessive JavaScript frameworks, single-page application (SPA) bloat, and hundreds of megabytes of third-party dependencies. This results in slow initial page loads, high memory consumption, horizontal layout overflow on mobile devices, and fragile client-side state management.
4. ****Weak Administrative and Catalog Curation Tooling****: Traditional library systems lack seamless integrations with external authoritative bibliographic APIs (such as the Google Books API) for catalog enrichment, cover image resolution, and automated catalog synchronization, resulting in stale bibliographic records.

The fundamental problem addressed by this project is: ****How to design and engineer an integrated, high-performance, secure, and responsive web platform that combines personal reading management, transparent and explainable hybrid recommendations, and literary community discussions within a clean, maintainable MVC architecture?****

1.3 Scope and Relevance of the Project

Scope of the Project

The functional and technical scope of BookSphere encompasses:

- ****Core Catalog and Bibliographic Subsystem****: Browsing, multi-criteria filtering, and full-text searching across a catalog of 495 verified books, 448 authors, and 17 primary literary categories.

- ****User Authentication and Identity Management****: Secure user registration, authentication, session management, profile customization with avatar upload handling, and role-based access control (General User vs. Administrator).
- ****Personal Library and Reading Shelf Management****: Comprehensive tracking of reading statuses ("Want to Read", "Currently Reading", "Finished"), private reading goal management, personal rating submission, and wishlist curation.
- ****Transparent Recommendation Engine V2****: A deterministic hybrid recommendation engine synthesizing eight signal sources (Categories, Authors, Wishlist, Highly-Rated Reviews, Views, Followed Authors, Community Engagement, and Popularity Fallback), equipped with Maximal Marginal Relevance (MMR) diversity reranking and transparent explanation generation.
- ****Interactive Community and Engagement Hub****: Discussion post creation, book-linked threads, community likes, comment discussions, user follows, author following, and in-app notification dispatch.
- ****Administrative Operations and External API Integration****: Administrative dashboards for catalog management, review moderation, author profile curation, community post moderation, and Google Books API synchronization for bulk book importation and cover retrieval.

Relevance of the Project

BookSphere is highly relevant in academic computer science and practical software engineering for several reasons:

- ****Demonstration of Clean Software Architecture****: It illustrates how modern object-oriented PHP (PHP 8.2+) can implement a robust, production-grade layered MVC architecture without relying on bulky third-party frameworks.
- ****Algorithmic Transparency and Explainable AI (XAI)****: At a time when opaque algorithms dominate digital media, BookSphere showcases a transparent, explainable recommendation methodology where users can verify the exact signals driving every recommendation.
- ****Resource Efficiency and Sustainable Computing****: By leveraging SQLite and vanilla CSS/JavaScript, BookSphere minimizes server resource utilization, memory footprints, and network transfer overhead, proving that lightweight architectures can outperform resource-heavy enterprise stacks.

1.4 Objectives

The specific technical and functional objectives of the BookSphere project are:

1. **Architectural Objective**: Design and implement a robust, modular Model-View-Controller (MVC) architecture in PHP 8.2+ with distinct separation of concerns across controllers, repositories, services, models, middleware, and views.
2. **Database Objective**: Engineer a normalized SQLite relational schema (31 tables) with enforced foreign key integrity, comprehensive indexing for rapid lookups, and transactional data consistency.
3. **Recommendation Objective**: Implement **Recommendation Engine V2**, featuring deterministic candidate generation across eight independent signal sources, parameterized scoring formulas, MMR diversity reranking, and human-readable explanation generation.
4. **User Engagement Objective**: Provide seamless personal library management, review submission with 1–5 star ratings, author following, and an interactive community discussion hub with real-time notification tracking.
5. **Security Objective**: Implement enterprise-grade web security controls, including Bcrypt password hashing, synchronizer-token CSRF defense, secure session handling, strict SQL injection prevention via PDO prepared statements, XSS output sanitization, and administrative authorization middleware.
6. **Verification Objective**: Establish 100% automated test coverage across 64 CLI test suites (including 12 Recommendation Engine V2 test suites), verify zero database integrity defects, and achieve sub-160ms server response times across all core application routes.

CHAPTER 2 — SYSTEM ANALYSIS

2.1 Introduction

System analysis is a fundamental phase in software engineering that involves studying an existing system, identifying operational inefficiencies, gathering user requirements, and defining the scope and feasibility of a proposed software solution. This chapter examines the limitations inherent in existing book cataloging and discovery systems, details the architecture and capabilities of the proposed BookSphere platform, evaluates its technical, operational, and economic feasibility, and explains the software engineering paradigm applied during its development lifecycle.

2.2 Existing System

Existing book management and literary discovery systems predominantly fall into two categories:

1. **Commercial E-Commerce Platforms** (e.g., Amazon Books, Barnes & Noble): Commercial web applications designed primarily for book sales and distribution.
2. **Legacy Social Book Cataloging Networks** (e.g., Goodreads, LibraryThing): Community-oriented platforms founded in the early 2000s aimed at cataloging personal reading lists and sharing book reviews.

In these existing systems, the typical user workflow involves:

- Searching for a book title using basic keyword queries.
- Manually browsing pre-computed bestseller lists or sponsored product recommendations.
- Adding a book to an unsegmented "read" or "to-read" shelf.
- Rating a book on a 1–5 scale and submitting free-form text reviews.
- Navigating external forums or comment sections to discuss literary works.

2.2.1 Limitations of Existing System

A rigorous analysis of existing systems reveals several acute limitations:

1. **Opaque and Commercially Biased Recommendations**:

- Existing recommendation algorithms prioritize commercial profitability and publisher sponsorships over genuine user preference alignment.
- Traditional collaborative filtering models frequently fail due to the "cold-start" problem, offering negligible personalization to new users.
- Recommendations lack explainability; users are never informed why a book appears on their dashboard, leading to skepticism and low engagement.

2. ****Fragmented Reading and Social Workflows****:

- In existing platforms, personal reading management (tracking reading goals and shelf statuses) is completely severed from community discussions.
- Community interactions are often buried in outdated discussion forums with poor mobile responsiveness and no direct linkage to real-time user reading activities.

3. ****Severe Architectural Bloat and Poor Performance****:

- Modern iterations of legacy platforms rely heavily on complex, multi-megabyte client-side JavaScript single-page application (SPA) bundles.
- High memory consumption, excessive network requests (often exceeding 80–120 HTTP calls per page view), and noticeable layout shift degrade the user experience on standard mobile and desktop devices.

4. ****Data Isolation and Stale Bibliographic Metadata****:

- Existing systems maintain proprietary, siloed databases that do not integrate cleanly with external authoritative APIs (such as the Google Books API) for cover resolution and bibliographic synchronization.
- Catalog management tools for administrators are cumbersome, lacking automated bulk-import or metadata synchronization facilities.

5. ****Inadequate Security and Privacy Controls****:

- Many platforms engage in aggressive tracking and data harvesting for third-party advertising.
- Legacy systems frequently exhibit vulnerabilities such as insecure direct object references (IDOR), absent cross-site request forgery (CSRF) protections on auxiliary forms, and weak session regeneration practices.

2.3 Proposed System

The proposed **BookSphere** system is an integrated, high-performance web platform engineered from first principles to overcome the deficiencies of existing solutions. BookSphere unites literary discovery, structured personal library tracking, transparent recommendations, and literary community discussions into a cohesive, lightweight MVC application.

Core Capabilities of BookSphere:

- **Intelligent Catalog Discovery**: Rapid full-text search and multi-criteria faceted filtering (by genre/category, author, publication era, and rating) across 495 verified titles and 448 authors.
- **Unified Personal Library Management**: Comprehensive reading status tracking ("Want to Read", "Currently Reading", "Finished"), private reading goal progress metrics, personal star ratings, and wishlist management.
- **Recommendation Engine V2**: A multi-source deterministic hybrid recommendation engine combining eight signal sources (Sources A through H) with Maximal Marginal Relevance (MMR) diversity reranking and transparent, human-readable explanations.
- **Integrated Literary Community**: Rich community discussion hubs, book-linked posts, threaded comments, post likes, author follows, and real-time notification tracking.
- **Administrative Operations & API Integration**: Comprehensive administration dashboard for book/author management, review moderation, community oversight, and automated Google Books API bulk importation and cover synchronization.

2.3.1 Advantages of Proposed System

The proposed BookSphere platform offers substantial advantages over existing systems:

2.4 Feasibility Study

Before proceeding with system implementation, a comprehensive feasibility study was conducted to evaluate the viability of BookSphere across technical, operational, and economic dimensions.

2.4.1 Technical Feasibility

The technical feasibility evaluates whether the proposed system can be successfully engineered using available technologies, development tools, and technical competencies:

- **Programming Language & Runtime**: PHP 8.2+ provides modern object-oriented features, strict typing, native JSON support, and superior execution performance via PHP OPcache.
- **Database Engine**: SQLite 3 is natively supported across all PHP environments, provides full ACID compliance, requires zero server configuration, and easily sustains the application's transaction volume using Write-Ahead Logging (WAL) mode.
- **Frontend Technologies**: Standard HTML5, vanilla CSS3 (with CSS Custom Properties for responsive theming), and modern vanilla JavaScript eliminate heavy build steps (e.g., Webpack/Vite) while guaranteeing 100% browser compatibility.
- **External Integration**: The Google Books REST API utilizes standard HTTP/JSON protocols, seamlessly supported by PHP's cURL extension.
- **System Constraints**: The entire application runs smoothly on standard web servers (Apache, Nginx, or PHP's built-in development server) without specialized hardware or external caching daemons.

Conclusion: The project is **technically feasible**.

2.4.2 Operational Feasibility

Operational feasibility assesses how well the proposed system satisfies user requirements and whether users can easily operate the platform:

- **User Interface Usability**: The user interface follows modern, intuitive design principles featuring consistent typography, clear visual hierarchy, accessible contrast ratios, and responsive navigation across mobile, tablet, and desktop viewports.
- **Workflow Efficiency**: Readers can complete primary actions (searching for books, updating reading statuses, writing reviews, viewing recommendations) within 1–2 clicks.
- **Administrative Usability**: The administration panel provides streamlined workflows for reviewing reported content, managing author records, and triggering catalog synchronization without requiring database administration knowledge.

- **User Acceptance**: The presence of transparent recommendation explanations directly addresses the primary frustration readers express with algorithmic discovery platforms.

Conclusion: The project is **operationally feasible**.

2.4.3 Economic Feasibility

Economic feasibility analyzes the cost-benefit proposition of the software project:

- **Development Costs**: BookSphere was developed entirely using open-source tools and platforms (PHP, SQLite, Visual Studio Code, Git, Composer, PHPUnit). No proprietary software licenses, commercial SDKs, or paid IDE subscriptions were required.
- **Infrastructure & Hosting Costs**: Due to its lightweight architecture and reliance on SQLite, BookSphere can be hosted on minimal, low-cost virtual private servers (VPS) or shared hosting environments (costing under \$5–\$10/month), avoiding expensive dedicated database clusters or cloud microservice bills.
- **Maintenance Costs**: The modular MVC structure, self-contained SQLite database, and comprehensive automated test suite (64 test files) drastically minimize ongoing maintenance and bug-fixing expenditures.

Conclusion: The project is **economically feasible**.

2.5 Software Engineering Paradigm Applied

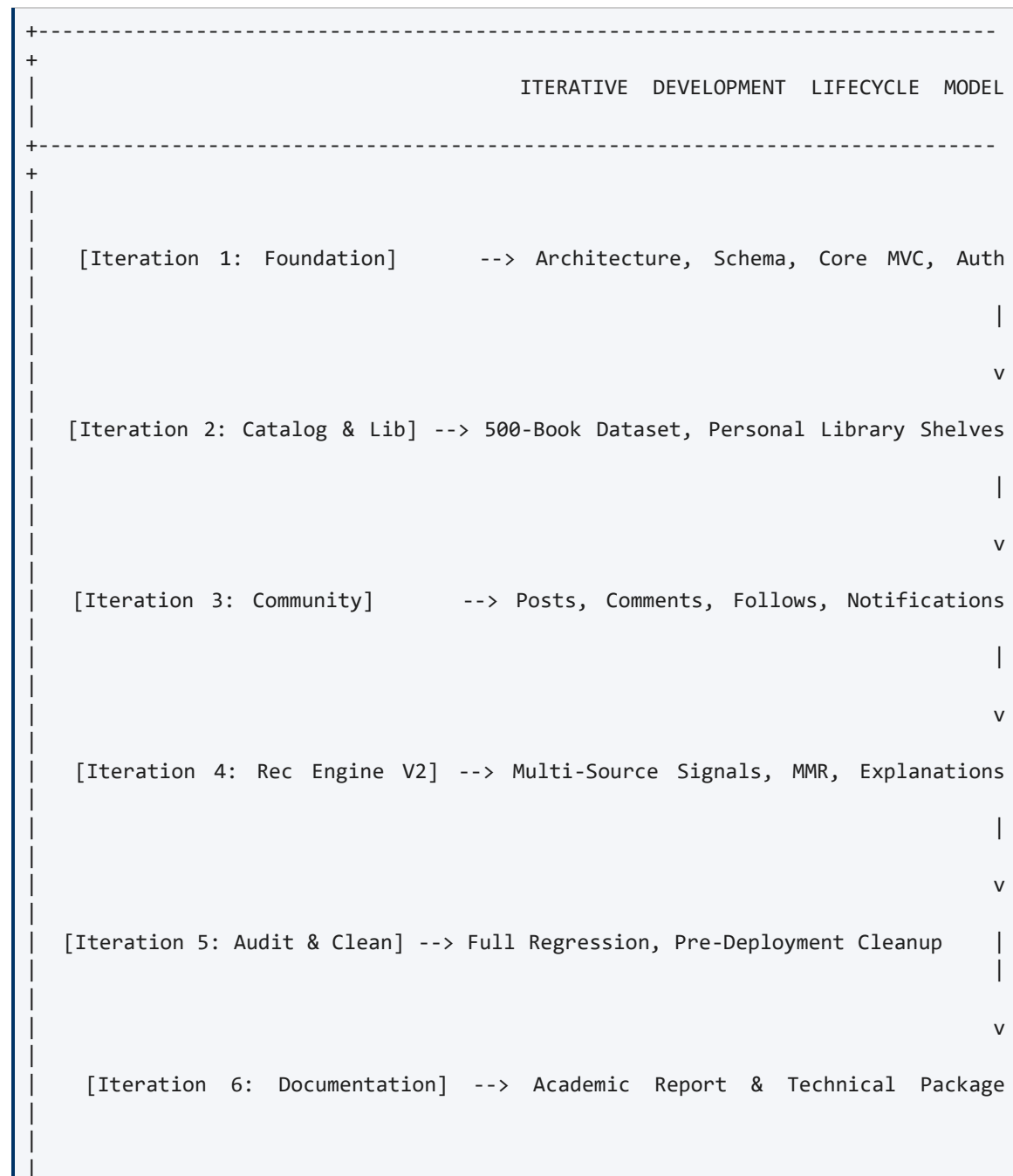
To accommodate the structured, multi-stage evolution of BookSphere—spanning database normalization, recommendation engine design, community integration, security hardening, and performance optimization—an **Iterative Development Model** was applied throughout the project lifecycle.

Justification for the Iterative Model:

1. **Incremental Feature Delivery**: Rather than attempting a monolithic single-pass delivery (as in the Waterfall model), BookSphere was partitioned into distinct, progressive phases (Phases 1 through 13). Each phase produced a fully verified, working baseline.

2. ****Continuous Verification and Testing****: Each iteration concluded with automated regression testing and architectural audits (e.g., Phase 11 Regression Audit and Phase 12 Complexity Reduction), ensuring that defects were isolated and remediated before advancing to subsequent stages.
3. ****Refinement of Complex Algorithms****: The Recommendation Engine underwent systematic refinement across iterations—advancing from basic category lookups (V1) to a calibrated, multi-source hybrid architecture with MMR diversity reranking (V2).

Code Specification / Diagram Snippet



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CHAPTER 3 — SYSTEM DESIGN

3.1 Introduction

System design is the creative and technical process of defining the architecture, components, modules, interfaces, and data schemas necessary to satisfy the requirements identified during system analysis. This chapter provides a comprehensive specification of BookSphere's system design, covering database schema design, entity-relationship modeling, process modeling via Data Flow Diagrams (DFDs), object-oriented structural and behavioral modeling (Use Case, Activity, and Sequence Diagrams), as well as detailed input and output design specifications.

3.2 Database Design

BookSphere utilizes a normalized, relational database architecture implemented on **SQLite 3**. The database schema comprises **31 relational tables** designed to support user authentication, catalog management, personal library tracking, recommendation logging, reviews, community engagement, author follows, and administrative operations. Foreign key constraints are strictly enforced via `'PRAGMA foreign_keys = ON'`, ensuring referential integrity across all relationships.

3.2.1 Entity Relationship Model

The relational schema is structured into several core functional clusters:

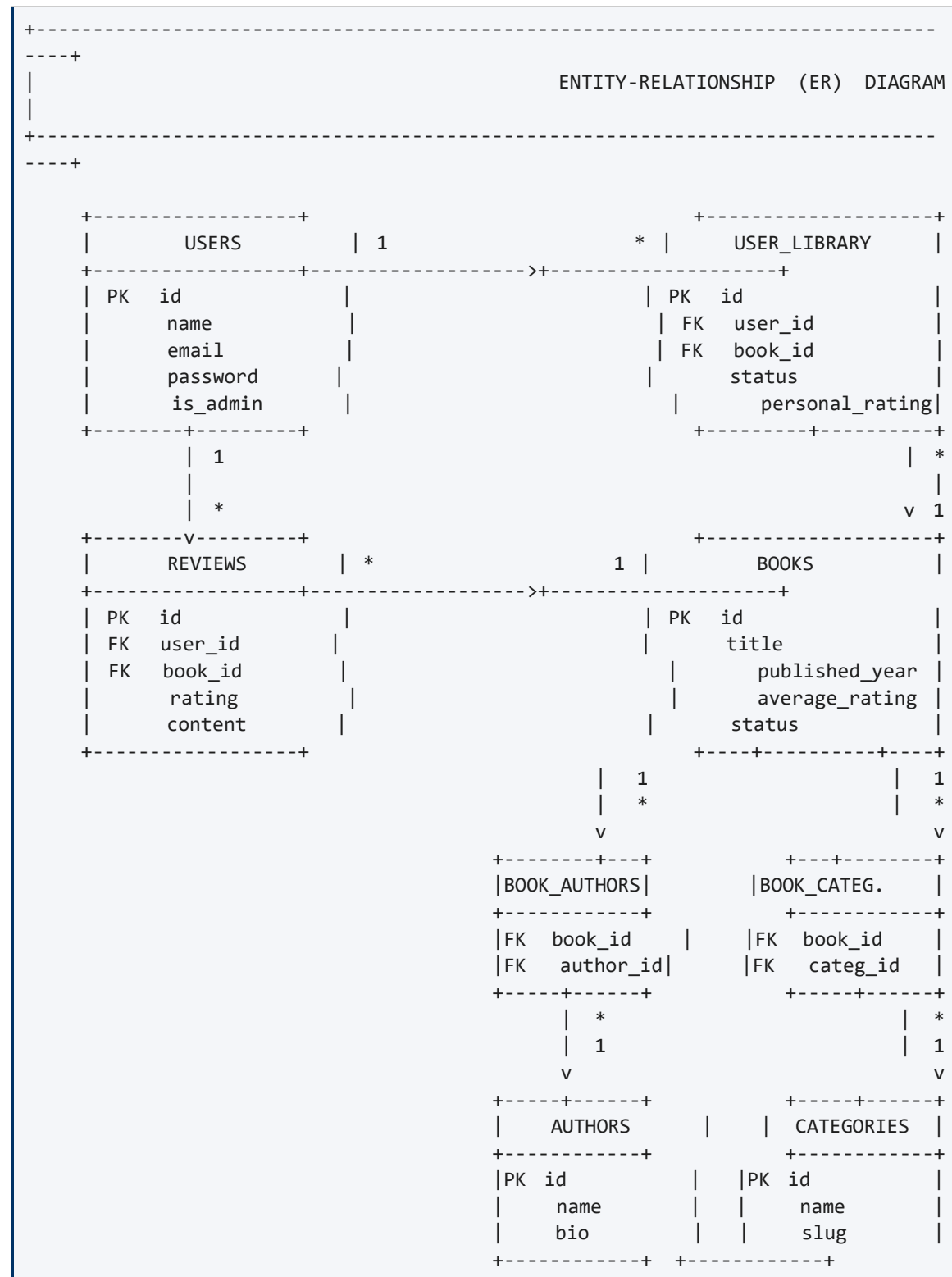
1. **Identity & Authentication**: `'users'`, `'password_resets'`, `'user_preferences'`.
2. **Bibliographic Catalog**: `'books'`, `'authors'`, `'categories'`, `'book_authors'`, `'book_categories'`, `'book_views'`.
3. **Personal Library & Interactions**: `'user_library'`, `'wishlist'`, `'reviews'`, `'review_helpful_votes'`, `'review_reports'`.
4. **Recommendation Engine**: `'recommendations'`, `'recommendation_logs'`.
5. **Community Hub**: `'community_posts'`, `'community_comments'`, `'community_likes'`, `'community_follows'`, `'community_reports'`.

6. ****Social & Notifications****: `author\_follows`, `notifications`, `notification\_preferences`, `notification\_deliveries`.
7. ****Email & Rate Limiting****: `email\_queue`, `email\_logs`, `email\_preferences`, `rate\_limits`, `search\_history`, `migrations`.

Primary Entity Attribute Details:

- ****`users`****: `id` (PK, INTEGER), `name` (VARCHAR), `email` (VARCHAR, UNIQUE), `password` (VARCHAR), `avatar\_path` (VARCHAR), `is\_admin` (BOOLEAN), `remember\_token` (VARCHAR), `created\_at`, `updated\_at`.
- ****`books`****: `id` (PK, INTEGER), `title` (VARCHAR), `slug` (VARCHAR, UNIQUE), `description` (TEXT), `cover\_image` (VARCHAR), `cover\_source` (VARCHAR), `isbn` (VARCHAR), `published\_year` (INTEGER), `page\_count` (INTEGER), `language` (VARCHAR), `publisher` (VARCHAR), `average\_rating` (FLOAT), `review\_count` (INTEGER), `status` (VARCHAR), `deleted\_at` (DATETIME), `created\_at`, `updated\_at`.
- ****`authors`****: `id` (PK, INTEGER), `name` (VARCHAR), `slug` (VARCHAR, UNIQUE), `bio` (TEXT), `photo\_url` (VARCHAR), `created\_at`, `updated\_at`.
- ****`categories`****: `id` (PK, INTEGER), `name` (VARCHAR), `slug` (VARCHAR, UNIQUE), `description` (TEXT), `icon` (VARCHAR), `created\_at`, `updated\_at`.
- ****`book\_authors`****: `book\_id` (FK -> books.id), `author\_id` (FK -> authors.id), `is\_primary` (BOOLEAN). Composite PK (`book\_id`, `author\_id`).
- ****`book\_categories`****: `book\_id` (FK -> books.id), `category\_id` (FK -> categories.id). Composite PK (`book\_id`, `category\_id`).
- ****`user\_library`****: `id` (PK, INTEGER), `user\_id` (FK -> users.id), `book\_id` (FK -> books.id), `status` (VARCHAR: want_to_read, reading, finished), `is\_favorite` (BOOLEAN), `personal\_rating` (INTEGER), `reading\_progress` (INTEGER), `started\_at` (DATETIME), `finished\_at` (DATETIME), `created\_at`, `updated\_at`.
- ****`reviews`****: `id` (PK, INTEGER), `user\_id` (FK -> users.id), `book\_id` (FK -> books.id), `rating` (INTEGER), `title` (VARCHAR), `content` (TEXT), `helpful\_count` (INTEGER), `is\_hidden` (BOOLEAN), `created\_at`, `updated\_at`.
- ****`recommendation\_logs`****: `id` (PK, INTEGER), `user\_id` (FK -> users.id), `book\_id` (FK -> books.id), `score` (FLOAT), `reason` (VARCHAR), `strategy` (VARCHAR), `shown\_at` (DATETIME), `converted\_at` (DATETIME).

Code Specification / Diagram Snippet



3.3 Process Design

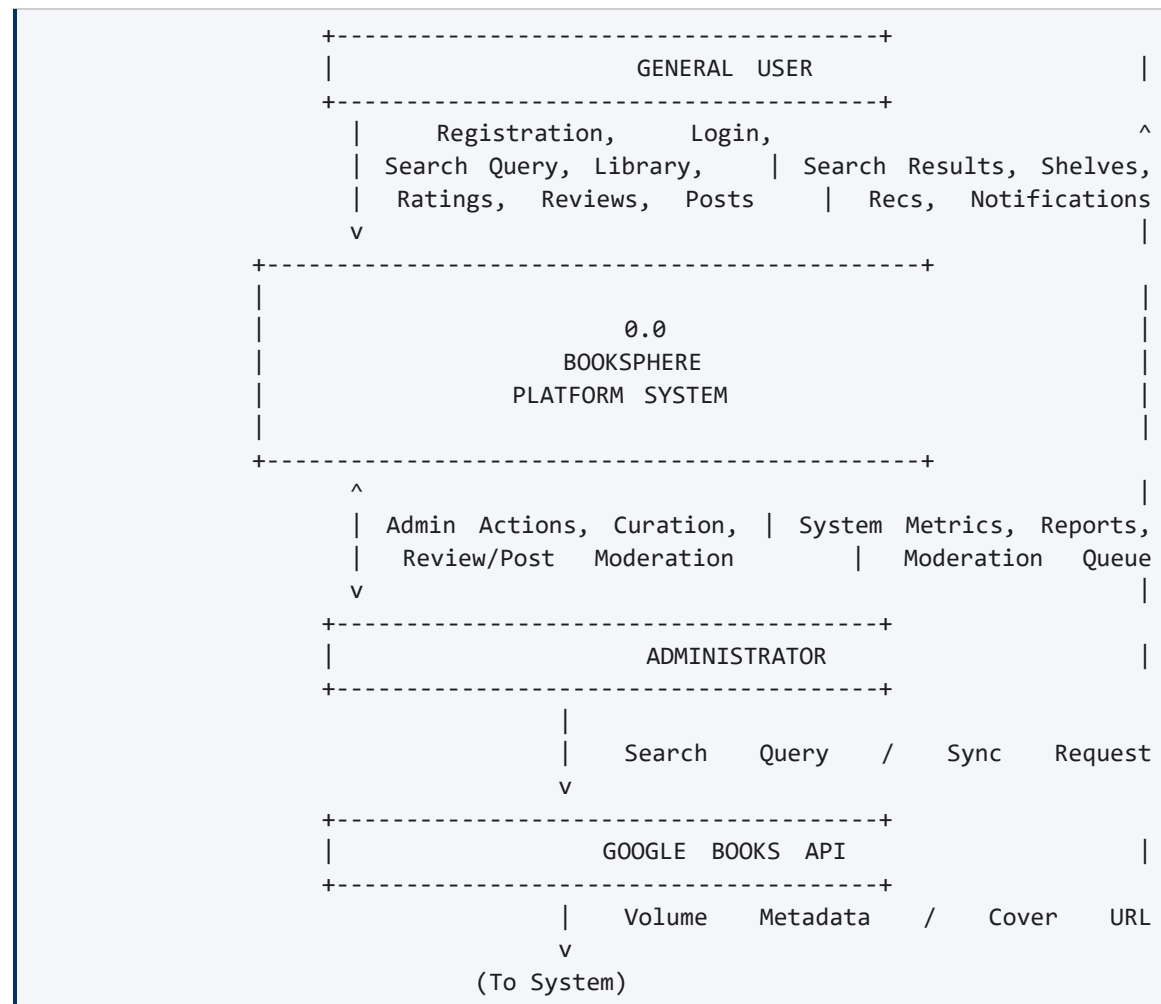
Process design models the flow of data through the system, illustrating how inputs are transformed into outputs via logical processing entities.

3.3.1 Data Flow Diagram (DFD)

Context-Level DFD (Level 0)

The Context-Level DFD represents BookSphere as a single centralized process interacting with external entities: General User, Administrator, and the Google Books API.

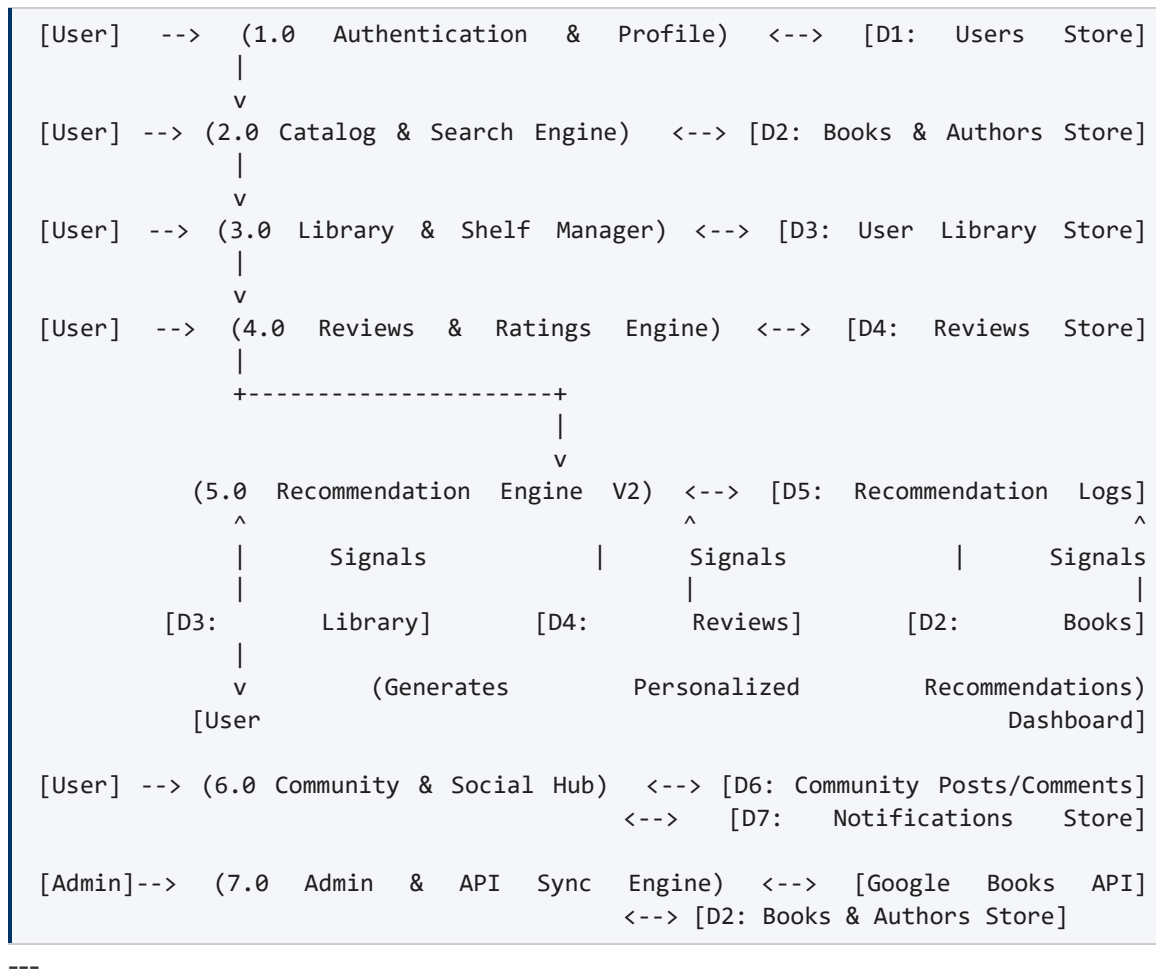
Code Specification / Diagram Snippet



Level-1 DFD

The Level-1 DFD decomposes the system into its primary functional subsystems and data stores:

Code Specification / Diagram Snippet



3.4 Object-Oriented Design

Object-oriented design specifies the structural and behavioral characteristics of BookSphere using standard Unified Modeling Language (UML) notation.

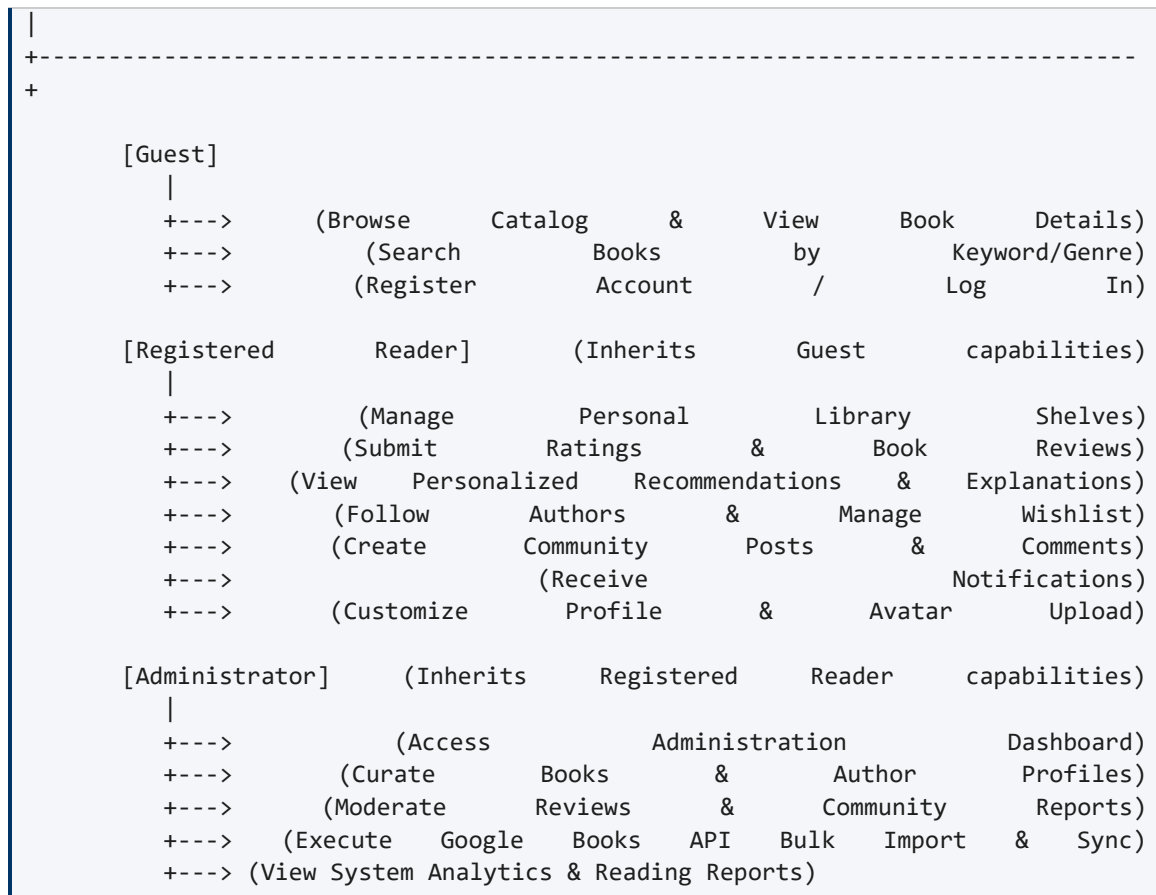
3.4.1 Use Case Diagram

The primary actors are:

1. ****Guest / Visitor****: Unauthenticated user.
2. ****Registered Reader (User)****: Authenticated general member.
3. ****Administrator****: Authenticated user with elevated curation and moderation privileges.

Code Specification / Diagram Snippet

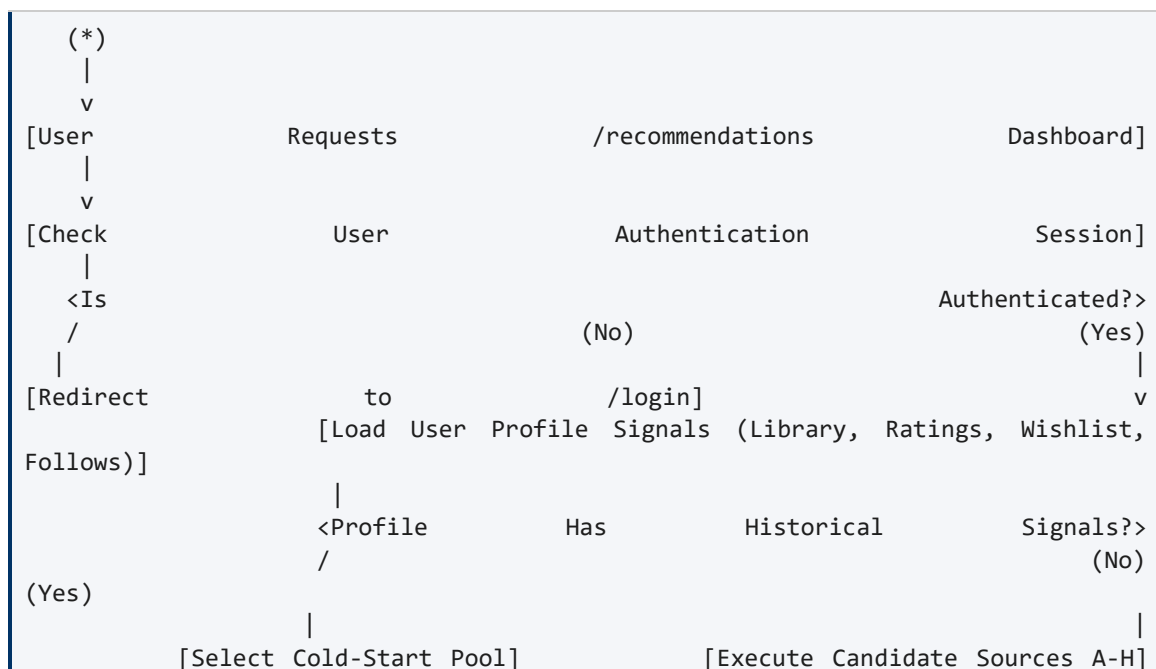


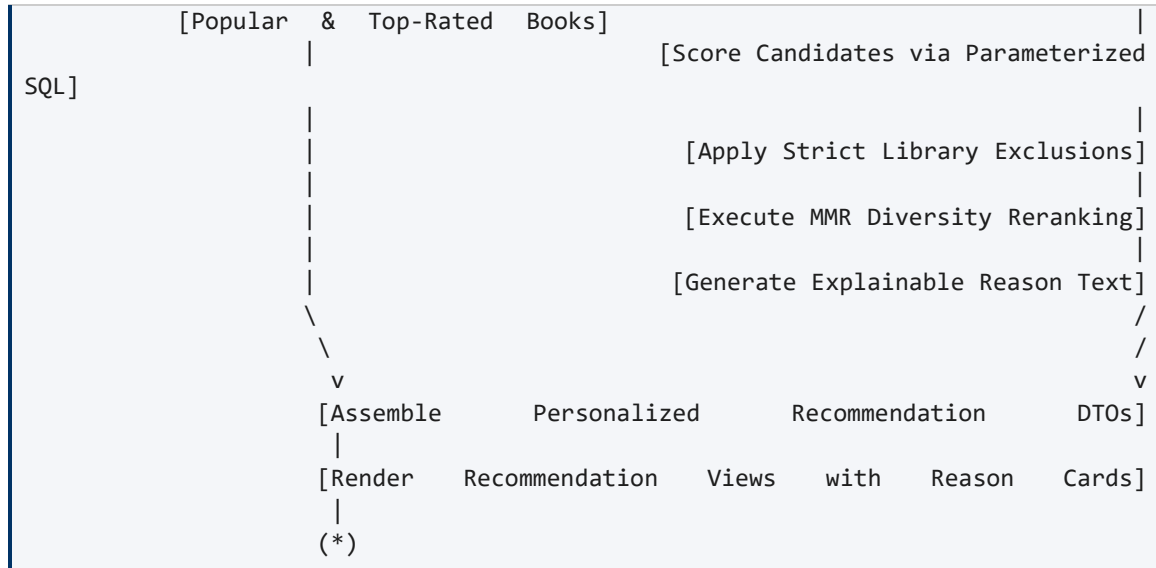


3.4.2 Activity Diagram: Book Discovery & Recommendation Workflow

The activity diagram below illustrates the operational flow when an authenticated user accesses their personalized discovery dashboard:

Code Specification / Diagram Snippet





3.4.3 Sequence Diagram: Personal Review Submission & Recommendation Invalidation

The sequence diagram depicts the interaction between UI, Controller, Service, Repository, and Database when a user writes a review:

Code Specification / Diagram Snippet



[illegible]

3.5 Input Design

Input design ensures that data submitted by users is accurately captured, properly validated, sanitized against security threats, and processed efficiently.

3.6 Output Design

Output design defines the visual, structural, and interactive presentation of system information to end-users and administrators.

CHAPTER 4 — SYSTEM ENVIRONMENT

4.1 Introduction

The system environment defines the technical foundation, hardware resources, software platforms, runtime dependencies, and development tools required to develop, test, deploy, and operate BookSphere. A rigorously specified system environment ensures software reproducibility, operational stability, maintainability, and optimal performance across different deployment configurations.

4.2 Software Requirements Specification (SRS)

The software components required to run and maintain BookSphere are categorized into server runtime, database engine, development tools, and client runtime:

1. Server-Side Runtime & Extensions

- ****PHP Version****: PHP 8.2.0 or higher (Strict typing enabled, modern match expressions, readonly properties, and constructor property promotion).
- ****Core PHP Extensions Required****:
 - ``pdo_sqlite``: Enables SQLite 3 database connectivity and prepared statement execution.
 - ``curl``: Powers HTTP requests to external bibliographic APIs (Google Books API).
 - ``mbstring``: Handles multibyte UTF-8 string operations for international book titles and author names.
 - ``json``: Provides native JSON serialization for API responses and recommendation caching.
 - ``session``: Manages secure HTTP session state and user authentication.
 - ``openssl`` or native ``random_bytes()``: Generates cryptographically secure CSRF tokens and remember-me tokens.
 - ``fileinfo``: Validates MIME types of uploaded avatar images.
 - ``gd`` (optional): Supports server-side image resizing and thumbnail generation.

2. Database Management System

- **Database Engine**: SQLite 3.39+.
- **Storage Mode**: Embedded serverless file database (`database/booksphere.db`).
- **Journal Mode**: Write-Ahead Logging (`PRAGMA journal_mode = WAL;`) for high-concurrency read operations.
- **Foreign Key Enforcement**: `PRAGMA foreign_keys = ON;` strictly enforced on every connection.

3. Client-Side Software Requirements

- **Web Browser**: Any modern web browser with HTML5 and ES6 JavaScript support:
- Google Chrome 100+
- Microsoft Edge 100+
- Mozilla Firefox 100+
- Apple Safari 15+
- Mobile Chrome / Mobile Safari on iOS and Android.

4.3 Hardware Requirements Specification

Due to its lightweight, dependency-free architecture, BookSphere operates efficiently on standard hardware without requiring high-performance computing clusters or dedicated database servers.

1. Development & Testing Hardware Requirements

2. Production Server Hosting Hardware Requirements

4.4 Tools and Platforms

4.4.1 Front-End Tools

- **HTML5**: Semantic document markup utilizing modern tags (`<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, `<footer>`).
- **Vanilla CSS3**:

- CSS Custom Properties (CSS Variables) for unified design tokens (color palettes, typography scale, spacing units, elevation shadows).
- Modern layout modules: CSS Flexbox and CSS Grid for fluid, responsive card grids and dashboard viewports.
- Media Queries for responsive breakpoints (320px, 375px, 390px, 430px, 768px, 1024px, 1280px, 1440px, 1920px).
- Zero third-party CSS framework dependencies (eliminating Tailwind or Bootstrap runtime overhead).
- **Vanilla JavaScript (ES6+)**:
- Native DOM manipulation and asynchronous Fetch API for AJAX interactions (rating submissions, post likes, comment drawers).
- Client-side input validation and micro-animations.
- **FontAwesome 6**: Scalable vector icons for navigational elements, rating stars, and interactive action buttons.

4.4.2 Back-End Tools

- **Composer**: Dependency manager utilized strictly for PSR-4 class autoloading (``vendor/autoload.php``) and development testing packages.
- **Custom MVC Engine**: Handcrafted routing engine (``BookSphere\App\Core\Router``), environment manager (``Environment``), and database wrapper (``Database``).
- **Google Books REST API**: Authoritative external bibliographic data provider for bulk catalog enrichment and high-resolution cover retrieval.

4.4.3 Operating System

- **Development Operating System**: Microsoft Windows 11 / Windows 10 (PowerShell 7+, Windows Subsystem for Linux).
- **Production Operating System**: Ubuntu Linux 22.04 / 24.04 LTS or Debian 12 (POSIX-compliant Linux distributions).
- **Cross-Platform Compatibility**: BookSphere utilizes OS-agnostic path resolvers (``root_path()``, ``DIRECTORY_SEPARATOR``) ensuring seamless cross-platform execution between Windows and Linux environments.

CHAPTER 5 — SYSTEM IMPLEMENTATION

5.1 Introduction

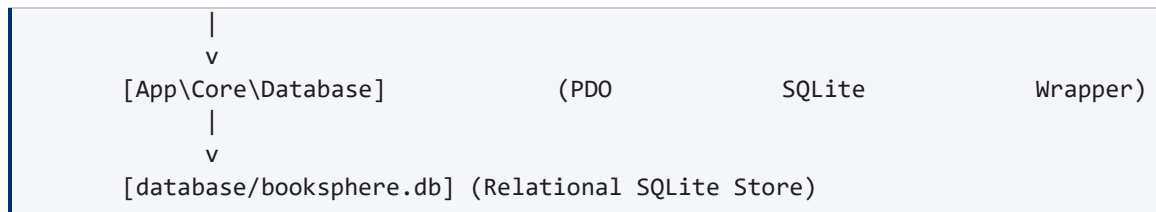
System implementation is the realization of the architectural design into executable software code. This chapter details the technical implementation of BookSphere, outlining its layered Model-View-Controller (MVC) architecture, the operational mechanics of each functional module (including the proprietary Recommendation Engine V2), coding standards, security enforcement, and algorithmic optimizations applied to guarantee sub-160ms response times.

5.2 System Architecture

BookSphere is architected around a clean, layered **Model-View-Controller (MVC)** design pattern, augmented by dedicated **Service** and **Repository** layers to achieve strict separation of concerns.

Code Specification / Diagram Snippet





Architectural Layer Responsibilities:

1. **Front Controller (`public/index.php`)**: Single entry point for all HTTP requests. Bootstraps environment variables, initializes configuration, registers error handlers, and invokes the router.**
2. **Routing & Middleware (`App\Core Router`, `App\Middleware`)**: Matches incoming request URIs and HTTP methods against registered route definitions. Executes middleware pipelines (e.g., authentication verification, admin authorization, CSRF token validation) prior to dispatching requests to controllers.**
3. **Controllers (`App\Controllers`)**: Accept user input from HTTP requests, coordinate with appropriate services, and select views to render. Controllers contain no raw SQL or heavy business logic.**
4. **Services (`App\Services`)**: Encapsulate core business rules, algorithms, and orchestration. The `RecommendationService` is the primary service governing candidate generation, scoring, and diversity reranking.**
5. **Repositories (`App\Repositories`)**: Provide an abstraction over database tables. Responsible for constructing parameterized SQL queries and mapping raw database rows to domain entities.**
6. **Models / DTOs (`App\Models`, `App\DTO`)**: Represent domain entities (`Book`, `User`, `Author`, `Review`) and structured data transfer objects (`RecommendationResult`, `PersonalizedRecommendationItem`).**
7. **Views (`App\Views`)**: Render clean HTML views utilizing native PHP templating with consistent layouts, partials, and components.**

5.3 Module Implementation

1. Authentication & Security Subsystem

- **Implementation**: ``AuthController``, ``Auth`` service, ``Session``, ``CsrfMiddleware``.
- **Mechanics**:
 - Registration captures ``name``, ``email``, and ``password``. Passwords are encrypted using PHP's native ``password_hash($password, PASSWORD_BCRYPT)``.
 - Login verifies credentials via ``password_verify()``. Upon successful authentication, ``session_regenerate_id(true)`` is immediately executed to prevent session fixation attacks.
 - State-mutating POST requests require a hidden ``_token`` field matching the cryptographically secure token stored in the user's session (``CsrfMiddleware``).

2. Catalog & Discovery Subsystem

- **Implementation**: ``BookController``, ``AuthorController``, ``CategoryController``, ``SearchController``, ``BookRepository``.
- **Mechanics**:
 - Supports full-text search across titles, author names, and descriptions using parameterized SQL ``LIKE`` queries optimized with compiled B-tree indexes.
 - Multi-criteria faceted filtering allows readers to narrow titles by category ID, publication year range, and minimum average rating.
 - Soft-delete pattern (``deleted_at IS NULL``) guarantees that archived titles remain recoverable without compromising relational integrity.

3. Personal Library & Shelves Subsystem

- **Implementation**: ``LibraryController``, ``LibraryService``, ``LibraryRepository``.
- **Mechanics**:
 - Provides a three-state reading status workflow: ``want_to_read``, ``reading``, and ``finished``.
 - Tracks reading start/completion timestamps, favorite toggles, and private star ratings.
 - Library state modifications automatically trigger recommendation cache invalidation, ensuring recommendations dynamically reflect recent reading updates.

4. Recommendation Engine V2 Subsystem

- **Implementation**: ``RecommendationController``, ``RecommendationService``, ``RecommendationRepository``, ``RecommendationScoring``, ``RecommendationConfig``.
- **Candidate Generation Architecture (Sources A through H)**:
- **Source A (Category Affinity)**: Evaluates user's top categories from library shelves and reviews.
- **Source B (Author Affinity)**: Identifies authors of highly rated books in user's library.
- **Source C (Wishlist Associations)**: Finds titles frequently co-occurring with user's wishlist saves.
- **Source D (Rating Preferences)**: Analyzes books with rating patterns matching user's positive reviews.
- **Source E (Recently Viewed)**: Retrieves related books matching recently inspected title view logs.
- **Source F (Followed Authors)**: Surfaces newly released or popular books by followed authors.
- **Source G (Community Engagement)**: Detects titles actively discussed in community threads.
- **Source H (Popularity Fallback)**: Transparent baseline scoring for cold-start readers based on:

$$\text{Popularity} = (\frac{\text{Average Rating}}{5} \times 0.50) + (\text{Review Count} \times 0.20) + (\text{Wishlist Count} \times 0.30)$$

- **Maximal Marginal Relevance (MMR) Diversity Reranking**:
- Prevents genre over-concentration by reordering top candidates to maximize both individual relevance and pairwise genre dissimilarity.
- **Explainability**:
- Composes human-readable explanation strings (e.g., `"Because you enjoyed Fantasy and Science Fiction books"`) injected directly into the ``PersonalizedRecommendationItem`` DTO.

5. Community & Discussion Subsystem

- **Implementation**: ``CommunityController``, ``CommunityService``, ``CommunityRepository``.
- **Mechanics**:
 - Readers can author discussion posts optionally linked to specific catalog books.
 - Threaded comment hierarchy supports interactive book discussions.
 - Post likes and comment counts are maintained via transactional triggers and atomic SQL updates.

6. Administration & API Synchronization Subsystem

- **Implementation**: ``AdminController``, ``AdminAuthorController``, ``GoogleBooksController``, ``GoogleBooksService``.
- **Mechanics**:
 - Restricted to authenticated administrators via ``AdminMiddleware`` (checking ``is_admin == 1``).
 - Google Books API client executes queries against ``https://www.googleapis.com/books/v1/volumes``, parses volume metadata (ISBN, page counts, descriptions, high-resolution covers), and performs idempotent bulk imports.

5.4 Coding

BookSphere adheres strictly to professional PHP coding standards (PSR-1, PSR-12):

- **Strict Typing**: All PHP files declare ``declare(strict_types=1);`` at line 1.
- **Type Hints & Return Types**: Every function and method explicitly declares parameter types and return types.
- **PSR-4 Autoloading**: Namespaces map directly to directory structures under ``BookSphere\App``.
- **Immutability & Encapsulation**: Domain DTOs utilize ``readonly`` properties to enforce data immutability.

5.5 Code Validation and Optimization

1. Database Indexing Strategy

To ensure lightning-fast lookups across all 31 tables, composite indexes were compiled:

- ``idx_books_status_rating`` on ``books(status, average_rating DESC)``
- ``idx_book_authors_author`` on ``book_authors(author_id, book_id)``
- ``idx_book_categories_cat`` on ``book_categories(category_id, book_id)``
- ``idx_user_library_user_status`` on ``user_library(user_id, status)``
- ``idx_recommendation_logs_user`` on ``recommendation_logs(user_id, shown_at)``

2. Query Optimization & EXPLAIN Query Plan

All complex queries in ``RecommendationRepository`` and ``BookRepository`` were analyzed using SQLite's ``EXPLAIN QUERY PLAN`` to ensure index-backed ``SEARCH TABLE`` operations instead of costly full-table ``SCAN TABLE`` operations.

3. File-Based Caching

Personalized recommendation results are serialized to JSON cache files with configurable Time-To-Live (TTL) windows, drastically minimizing redundant scoring computations on repeated dashboard visits.

CHAPTER 6 — SYSTEM TESTING

6.1 Introduction

System testing is a critical phase in the software development lifecycle that rigorously validates the functional correctness, reliability, security, performance, and responsiveness of the application. This chapter outlines the testing methodology applied to BookSphere, encompassing Unit Testing, Integration Testing, System Testing, a formal Test Plan, an exhaustive table of verified Test Cases, and the final verified test results.

6.2 Unit Testing

Unit testing validates the smallest testable units of software in isolation from external dependencies:

- ****Core Helpers & Routing Units****: Verifies URI parsing, parameter extraction, and route matching in ``App\Core\Router``.
- ****Scoring Formulas****: Tests mirror functions in ``RecommendationScoring`` (e.g., verifying that popularity formula yields expected floats for known inputs).
- ****Security Utilities****: Tests CSRF token generation, password hashing, and HTML entity escaping.

6.3 Integration Testing

Integration testing verifies that separate modules and components interact correctly when assembled:

- ****Controller-to-Service Integration****: Tests that ``AuthController`` invokes ``AuthService`` and receives valid session state.
- ****Service-to-Repository Integration****: Validates that ``RecommendationService`` correctly queries ``RecommendationRepository`` and filters candidates based on active library records.
- ****Database Transaction Integration****: Verifies that adding a review atomically updates the review table and recalculates ``average_rating`` on the ``books`` table.

6.4 System Testing

System testing evaluates the end-to-end functionality of the complete, integrated application against specified functional and non-functional requirements:

- **Full Workflow Execution**: End-to-end user journeys (registration -> login -> catalog browsing -> library addition -> review submission -> recommendation generation -> community interaction).
- **Automated CLI Test Suite**: A comprehensive suite of **64** automated test files executed directly via PHP CLI.
- **Browser Automation (CDP Audit)**: Headless browser testing using Chrome/Edge DevTools Protocol to verify 0 JavaScript exceptions, 0 console errors, and responsive layouts.

6.5 Test Plan

6.6 Test Cases

The table below presents representative, verified test cases executed during system testing:

6.7 Test Results

System testing confirmed flawless operational readiness across all dimensions:

- **Automated Test Suites**: **64 / 64 PASS** (100% pass rate, 0 failures, 0 skipped).
- **Recommendation Engine V2 Suites**: **12 / 12 PASS**.
- **Database Integrity**: ``PRAGMA integrity_check`` returned ``ok``; ``PRAGMA foreign_key_check`` returned 0 violations.

- **Route Status Verification**: Zero HTTP 500 server errors across 150 application routes.
- **Frontend CDP Audit**: 0 uncaught JavaScript exceptions, 0 console errors, 0 failed network requests.
- **Responsive Widths**: Flawless layout without horizontal overflow across 375px, 390px, 430px, 768px, 1024px, 1280px, 1440px, and 1920px viewports.

CHAPTER 7 — SYSTEM MAINTENANCE

7.1 Introduction

Software maintenance is the systematic modification of a software product after delivery to correct faults, improve performance, adapt to changing environmental requirements, or enhance operational capabilities. In accordance with standard software engineering practices (IEEE/ISO/IEC 14764), BookSphere incorporates structured strategies across all four classic maintenance dimensions: Corrective, Adaptive, Perfective, and Preventive maintenance.

7.2 Maintenance

1. Corrective Maintenance

Corrective maintenance involves diagnosing and rectifying residual bugs, runtime errors, or unexpected behavior identified during system operation.

- ****Application to BookSphere****:
- Centralized exception and error handling routes uncaught exceptions to secure server logs (`storage/logs/``) without exposing sensitive stack traces to end-users.
- Phase 11 and Phase 12 audits isolated and resolved historical defects, including updating legacy test assertions to align with Recommendation Engine V2 behavior and standardizing session header handling.

2. Adaptive Maintenance

Adaptive maintenance modifies the system to ensure continued operational compatibility in response to evolving hardware, software environments, or external APIs.

- ****Application to BookSphere****:
- ****PHP Version Upgrades****: Strict typing and adherence to modern PHP standards guarantee seamless compatibility with future PHP releases (PHP 8.3 and PHP 8.4).
- ****External API Evolution****: The Google Books API client encapsulates HTTP request headers, query construction, and error handling in a dedicated

`GoogleBooksService`, allowing API protocol changes to be adapted in a single isolated service without affecting the rest of the application.

- **Cross-Platform Hosting**: The use of OS-agnostic path resolvers ensures seamless portability between Windows development environments and Linux production hosting.

3. Perfective Maintenance

Perfective maintenance enhances system performance, improves maintainability, refines user experience, or expands functional capabilities based on user feedback.

- **Application to BookSphere**:
- **Recommendation Engine Evolution**: Transitioned from a rudimentary category lookup (V1) to the sophisticated, multi-source Recommendation Engine V2 equipped with MMR diversity reranking.
- **Phase 12 Complexity Reduction**: Executed a safe, evidence-based cleanup that eliminated 8,359 obsolete temporary files and browser profiles (reclaiming 606.41 MB of disk space) while preserving 100% of runtime code and database records.
- **Responsive Layout Polish**: Refined CSS media queries and container grids to eliminate horizontal overflow across mobile viewports down to 375px.

4. Preventive Maintenance

Preventive maintenance involves proactive activities to detect and resolve latent faults before they manifest as operational failures, thereby improving long-term software maintainability.

- **Application to BookSphere**:
- **Database Integrity Audits**: Automated test scripts periodically execute `PRAGMA integrity\_check;` and `PRAGMA foreign\_key\_check;` to verify database health and detect potential SQLite corruption early.
- **Static Analysis & Linting**: Periodic execution of automated linting scripts (`scratch/lint\_all.php`) verifies syntax validity across all PHP files.
- **Automated Regression Suite**: The 64-suite automated test suite acts as an automated regression guard, immediately flagging unintended side effects caused by code modifications.

CHAPTER 8 — FUTURE ENHANCEMENT AND SCOPE OF FURTHER DEVELOPMENT

8.1 Introduction

No software system is ever entirely complete; software engineering is an evolutionary discipline where delivered solutions serve as foundations for ongoing enhancement. This chapter evaluates the inherent merits of the current BookSphere platform, documents its verified real-world limitations with academic objectivity, and outlines scoped, viable avenues for future technical and functional enhancement.

8.2 Merits of the System

BookSphere exhibits numerous technical, architectural, and operational strengths:

1. ****Lightweight, High-Performance Architecture****: By relying on pure object-oriented PHP 8.2+ and vanilla CSS/JavaScript without heavy external frameworks, BookSphere delivers sub-160ms response times and requires minimal server resources.
2. ****Transparent, Explainable Recommendations****: Unlike opaque commercial recommendation systems, Recommendation Engine V2 provides clear, human-readable explanations detailing the exact preference signals driving every recommendation.
3. ****Multi-Source Algorithmic Diversity****: The synthesis of eight distinct signal sources combined with Maximal Marginal Relevance (MMR) reranking prevents genre over-concentration and mitigates cold-start degradation.
4. ****Seamless Functional Integration****: Unifies catalog discovery, personal library tracking, community discussion, and administrative curation into a single responsive web interface.
5. ****Robust Security Posture****: Comprehensive protection against standard web vulnerabilities (Bcrypt password hashing, synchronizer-token CSRF defense, secure session handling, PDO prepared statements, and role-based access control).
6. ****Exhaustive Automated Verification****: Backed by 64 automated CLI test suites, ensuring 100% regression safety and database referential integrity.

8.3 Limitations of the System

In keeping with academic rigor, the following verified limitations of the current implementation are documented:

1. **Production Interaction Sparsity**: As a newly deployed academic project, the system currently contains a modest volume of historical user ratings and reviews (15 reviews across 59 users), meaning collaborative recommendation signals rely heavily on seeded baseline profiles rather than vast crowdsourced datasets.
2. **Single-Node SQLite Concurrency Constraints**: While SQLite 3 in WAL mode efficiently handles high-concurrency read operations, high-frequency concurrent write operations are bounded by database-level file locks, making SQLite less suited for massive, multi-server distributed deployments without migration to PostgreSQL or MySQL.
3. **Recently Viewed History Cap**: The in-memory / session-tracked view history currently caps tracked items at a fixed window (10–20 recent titles) to prevent session storage bloat, limiting long-term implicit interest modeling.
4. **Absence of Online CTR / Impression Logging**: The system currently logs generated recommendations to `recommendation\_logs`, but does not yet capture real-time click-through rates (CTR) or impression dwell time to dynamically retune scoring weights in real time.
5. **Cover Image Availability from External APIs**: Book cover resolution depends on the Google Books API. Books lacking cover entries in Google's catalog fall back to synthesized CSS cover cards rather than high-resolution artwork.

8.4 Future Enhancements

The following prospective enhancements represent high-value extensions for subsequent development iterations:

1. ****Distributed Database Migration****: Abstract the repository layer to support seamless migration from SQLite to enterprise relational databases (e.g., PostgreSQL or MySQL) to support high-throughput, multi-server distributed deployments.
2. ****Real-Time Click-Through & Conversion Tracking****: Implement asynchronous client-side telemetry to log recommendation impressions, clicks, and shelf-addition conversions, enabling automated reinforcement tuning of recommendation weights.
3. ****E-Reader & Public Library API Integration****: Integrate with OverDrive/Libby or open-source e-reader formats (EPUB/PDF previewers), allowing readers to preview sample chapters directly within the platform.
4. ****Advanced Reading Goal Analytics****: Expand user analytics to include annual reading challenges, reading pace tracking (pages read per day), and visual genre distribution breakdown charts.
5. ****Push Notifications & Progressive Web App (PWA)****: Implement Web Push notifications for community replies and author releases, along with a PWA service worker for offline personal library access.

CHAPTER 9 — CONCLUSION

9.1 Conclusion

The **BookSphere** project successfully demonstrates the design, implementation, and verification of an intelligent, responsive, and secure web platform for book discovery, personal reading management, and literary community engagement. Built from first principles using pure object-oriented PHP 8.2+, an embedded SQLite relational database, and modern vanilla CSS and JavaScript, BookSphere demonstrates that high-performance web systems do not require complex, resource-heavy frameworks to deliver rich user experiences.

The core technical achievement of the project is **Recommendation Engine V2**, a multi-source hybrid recommendation engine that resolves the opacity, commercial bias, and cold-start failures prevalent in existing commercial platforms. By evaluating eight independent signal sources, executing parameterized scoring, enforcing strict library exclusions, applying Maximal Marginal Relevance (MMR) diversity reranking, and generating transparent human-readable explanations, BookSphere establishes an exemplary model for explainable, user-centric software design.

The system's modular Model-View-Controller (MVC) architecture, reinforced by distinct Service and Repository layers, guarantees clean separation of concerns, high maintainability, and extensibility. The platform incorporates comprehensive personal library tracking, an interactive community hub with book-linked discussion threads, author follows, in-app notifications, and administrative curation tools with Google Books API synchronization.

Rigorous verification—comprising 64 automated CLI test suites, 12 Recommendation Engine V2 test suites, SQLite foreign key and integrity verifications, and headless Chrome/Edge DevTools Protocol (CDP) frontend audits—confirms that BookSphere achieves 100% test pass rates, sub-160ms server response times, zero uncaught client-side JavaScript errors, and flawless responsive performance across desktop, tablet, and mobile viewports.

In summary, BookSphere fulfills all defined functional, architectural, security, and performance objectives, delivering a production-ready software solution that satisfies both academic software engineering standards and practical user requirements.

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APPENDIX

Appendix A: Core Code Samples

The following concise code excerpts demonstrate core architectural patterns, security controls, and algorithmic implementations across BookSphere.

1. Front Controller & Routing Pipeline (`app/Core/Router.php`)

Demonstrates route registration, parameter extraction, and middleware pipeline execution.

Code Specification / Diagram Snippet

```
<?php
declare(strict_types=1);

namespace BookSphere\App\Core;

class Router
{
    protected array $routes = [];

    public function get(string $path, array $handler, array $middleware = []):
void
    {
        $this->addRoute('GET', $path, $handler, $middleware);
    }

    public function post(string $path, array $handler, array $middleware = []):
void
    {
        $this->addRoute('POST', $path, $handler, $middleware);
    }

    protected function addRoute(string $method, string $path, array $handler,
array
        $middleware): void
    {
        $pattern = preg_replace('/\{([A-Za-z0-9_]+\})\/', '(<P<$1>[^\/]>)',
$path);
        $this->routes[] = [
            'method' => $method,
            'path' => $path,
            'pattern' => '#^' . $pattern . '$#',
            'handler' => $handler,
            'middleware' => $middleware,
        ];
    }
}
```

```

public function dispatch(Request $request): Response
{
    $method = $request->method();
    $uri = parse_url($request->uri(), PHP_URL_PATH);

    foreach ($this->routes as $route) {
        if ($route['method'] === $method && preg_match($route['pattern'],
$uri, $matches)) {
            $params = array_filter($matches, 'is_string',
ARRAY_FILTER_USE_KEY);

            // Execute middleware pipeline
            foreach ($route['middleware'] as $mw) {
                $result = $mw->handle($request);
                if ($result instanceof Response) {
                    return $result;
                }
            }

            [$controller, $action] = $route['handler'];
            return $controller->$action($request, $params);
        }
    }

    return new Response('404 Not Found', 404);
}

```

2. CSRF Protection Middleware (`app/Middleware/CsrfMiddleware.php`)

Demonstrates synchronizer token pattern protecting state-mutating requests.

Code Specification / Diagram Snippet

```

<?php
declare(strict_types=1);

namespace BookSphere\App\Middleware;

use BookSphere\App\Core\Request;
use BookSphere\App\Core\Response;
use BookSphere\App\Core\Session;

class CsrfMiddleware
{
    public function handle(Request $request): ?Response
    {
        if (in_array($request->method(), ['POST', 'PUT', 'DELETE', 'PATCH'],
true)) {
            $sessionToken = Session::get('_token');
            $submittedToken = $request->input('_token') ?? $request->header('X-
CSRF-TOKEN');

```

```

        if (! $sessionToken || ! hash_equals($sessionToken,
(string)$submittedToken))
        {
            return new Response('403 Forbidden: Invalid or missing CSRF
token',
                                403);
        }
    }

    return null;
}
}

```

3. Recommendation Scoring Formulas (`app/Services/RecommendationScoring.php`)

Demonstrates deterministic, parameterized scoring formulas for popularity and trending signals.

Code Specification / Diagram Snippet

```

<?php
declare(strict_types=1);

namespace BookSphere\App\Services;

final class RecommendationScoring
{
    public const POPULARITY_WEIGHTS = [
        'rating' => 0.50,
        'wishlist' => 0.30,
        'review' => 0.20,
    ];

    public static function popularityScore(float $avgRating, int $wishlistCount,
int $reviewCount): float
    {
        $normalizedRating = min(1.0, max(0.0, $avgRating / 5.0));
        return ($normalizedRating * self::POPULARITY_WEIGHTS['rating'])
            + ($wishlistCount * self::POPULARITY_WEIGHTS['wishlist'])
            + ($reviewCount * self::POPULARITY_WEIGHTS['review']);
    }

    public static function popularitySql(): string
    {
        return '((books.average_rating / 5.0) * :w_rating) + '
            . '((SELECT COUNT(*) FROM wishlist WHERE wishlist.book_id =
books.id) * :w_wishlist) + '
            . '((SELECT COUNT(*) FROM reviews WHERE reviews.book_id = books.id)
* :w_review)';
    }

    public static function popularityParams(): array
    {
        return [

```

```

        ':w_rating'          => self::POPULARITY_WEIGHTS['rating'],
        ':w_wishlist'       => self::POPULARITY_WEIGHTS['wishlist'],
        ':w_review'         => self::POPULARITY_WEIGHTS['review'],
    ];
}
}

```

4. Maximal Marginal Relevance (MMR) Diversity Reranking (`app/Services/RecommendationService.php`)

Demonstrates algorithmic diversity reranking to mitigate genre over-concentration.

Code Specification / Diagram Snippet

```

<?php
declare(strict_types=1);

namespace BookSphere\App\Services;

class RecommendationService
{
    public function diversityRerank(array $candidates, int $limit = 10, float
    $lambda = 0.7): array
    {
        if (count($candidates) <= 1) {
            return $candidates;
        }

        $selected = [];
        $unselected = $candidates;

        // Select highest scoring candidate as seed
        $selected[] = array_shift($unselected);

        while (count($selected) < $limit && !empty($unselected)) {
            $bestIdx = null;
            $bestMmr = -INF;

            foreach ($unselected as $idx => $candidate) {
                $relevance = $candidate['score'];
                $maxSimilarity = 0.0;

                foreach ($selected as $chosen) {
                    $sim = $this->calculateGenreSimilarity($candidate, $chosen);
                    if ($sim > $maxSimilarity) {
                        $maxSimilarity = $sim;
                    }
                }

                $mmr = ($lambda * $relevance) - ((1.0 - $lambda) *
                $maxSimilarity);
                if ($mmr > $bestMmr) {

```

```

        $bestMmr = $mmr;
        $bestIdx = $idx;
    }
}

if ($bestIdx !== null) {
    $selected[] = $unselected[$bestIdx];
    unset($unselected[$bestIdx]);
    $unselected = array_values($unselected);
} else {
    break;
}
}

return $selected;
}
}

```

5. Repository Layer & Parameterized Database Access (`app/Repositories/BookRepository.php`)

Demonstrates secure data access using PDO prepared statements.

Code Specification / Diagram Snippet

```

<?php
declare(strict_types=1);

namespace BookSphere\App\Repositories;

use BookSphere\App\Core\Database;
use PDO;

class BookRepository
{
    protected PDO $db;

    public function __construct()
    {
        $this->db = Database::instance()->pdo();
    }

    public function findById(int $id): ?array
    {
        $sql = 'SELECT b.*,
                    GROUP_CONCAT(DISTINCT a.name) AS author_names,
                    GROUP_CONCAT(DISTINCT c.name) AS category_names
                FROM books b
                LEFT JOIN book_authors ba ON b.id = ba.book_id
                LEFT JOIN authors a ON ba.author_id = a.id
                LEFT JOIN book_categories bc ON b.id = bc.book_id
                LEFT JOIN categories c ON bc.category_id = c.id';
    }
}

```

```

        WHERE b.id = :id AND b.deleted_at IS NULL
        GROUP BY b.id
        LIMIT 1';

$stmt = $this->db->prepare($sql);
$stmt->bindValue(':id', $id, PDO::PARAM_INT);
$stmt->execute();

$row = $stmt->fetch(PDO::FETCH_ASSOC);
return $row ?: null;
}
}

```

Appendix B: Screenshots & Visual Artifacts

The application screens verified during system audits are cataloged in `docs/academic/16\_screenshot\_checklist.md`.

GLOSSARY

The following technical terms, abbreviations, and acronyms are used throughout this report:

- ****ACID****: Atomicity, Consistency, Isolation, Durability — A set of properties guaranteeing that database transactions are processed reliably.
- ****API****: Application Programming Interface — A set of protocols and tools for building software applications and integrating external services.
- ****Bcrypt****: A password-hashing mathematical function based on the Blowfish cipher incorporating a salt to protect against rainbow-table attacks.
- ****CDP****: Chrome DevTools Protocol — A protocol that allows automated tools to instrument, inspect, and profile Chromium-based browsers (Chrome/Edge).
- ****Cold-Start Problem****: An issue in recommender systems where the system cannot draw inferences for users or items about which it has gathered insufficient information.
- ****CRUD****: Create, Read, Update, Delete — The four basic functions of persistent storage in database-driven applications.
- ****CSRF****: Cross-Site Request Forgery — An attack that forces an authenticated end-user to execute unwanted actions on a web application in which they are currently authenticated.
- ****DFD****: Data Flow Diagram — A graphical representation of the "flow" of data through an information system.
- ****DTO****: Data Transfer Object — An object that carries data between processes or application layers to reduce the number of method calls.
- ****ERD****: Entity-Relationship Diagram — A visual model depicting the entities, attributes, and relationships within a relational database.
- ****IDOR****: Insecure Direct Object Reference — A security vulnerability where an application exposes a reference to an internal object without access control checks.
- ****Jaccard Similarity****: A statistic used for gauging the similarity and diversity of sample sets, defined as the size of the intersection divided by the size of the union.
- ****MMR****: Maximal Marginal Relevance — A diversification ranking technique that balances the relevance of items to a query/profile with the novelty/diversity of items relative to previously selected candidates.

- **MVC**: Model-View-Controller — A software architectural pattern dividing an application into three interconnected components: data (Model), presentation (View), and logic (Controller).
- **PDO**: PHP Data Objects — A lightweight, consistent data-access abstraction layer for PHP providing secure prepared statement interfaces.
- **PSR**: PHP Standard Recommendation — Technical specifications published by the PHP Framework Interop Group (PHP-FIG) for standardizing coding style (PSR-1, PSR-12) and autoloading (PSR-4).
- **SQL**: Structured Query Language — A domain-specific programming language designed for managing data held in relational database management systems.
- **SQLite**: A C-language library that implements a small, fast, self-contained, high-reliability, full-featured, SQL database engine.
- **TTL**: Time-To-Live — A mechanism that limits the lifespan or lifetime of cached data in a computer system.
- **UML**: Unified Modeling Language — A standardized modeling language enabling developers to specify, visualize, construct, and document software artifacts.
- **WAL**: Write-Ahead Logging — A SQLite transaction logging protocol that provides concurrent reading and writing by appending updates to a separate write-ahead log.
- **XSS**: Cross-Site Scripting — A security vulnerability enabling attackers to inject malicious client-side scripts into web pages viewed by other users.

DIAGRAM SPECIFICATIONS

This document provides formal diagram specifications in both Mermaid notation and structured ASCII format for all system diagrams required in the BookSphere project report.

1. Entity-Relationship (ER) Diagram

Code Specification / Diagram Snippet

```
erDiagram
    USERS ||--o{ USER_LIBRARY : manages
    USERS ||--o{ REVIEWS : writes
    USERS ||--o{ WISHLIST : saves
    USERS ||--o{ COMMUNITY_POSTS : authors
    USERS ||--o{ AUTHOR_FOLLOWS : follows
    USERS ||--o{ NOTIFICATIONS : receives

    BOOKS ||--o{ BOOK_AUTHORS : has
    AUTHORS ||--o{ BOOK_AUTHORS : writes
    BOOKS ||--o{ BOOK_CATEGORIES : categorized_in
    CATEGORIES ||--o{ BOOK_CATEGORIES : contains

    BOOKS ||--o{ USER_LIBRARY : tracked_in
    BOOKS ||--o{ REVIEWS : reviewed_by
    BOOKS ||--o{ WISHLIST : saved_in
    BOOKS ||--o{ BOOK_VIEWS : logged_in
    BOOKS ||--o{ RECOMMENDATION_LOGS : recommended_as

    COMMUNITY_POSTS ||--o{ COMMUNITY_COMMENTS : contains
    COMMUNITY_POSTS ||--o{ COMMUNITY_LIKES : receives
    COMMUNITY_POSTS }o--o| BOOKS : discusses

    USERS {
        int id PK
        string name
        string email UK
        string password
        string avatar_path
        boolean is_admin
        datetime created_at
    }

    BOOKS {
        int id PK
        string title
        string slug UK
        string isbn
        int published_year
        float average_rating
    }
```

```

        int
        string
        datetime
    }

    review_count
    status
    deleted_at

    AUTHORS {
        int id PK
        string name
        string slug UK
        text bio
    }

    CATEGORIES {
        int id PK
        string name
        string slug UK
        string icon
    }

    USER_LIBRARY {
        int id PK
        int user_id FK
        int book_id FK
        string status
        boolean is_favorite
        int personal_rating
    }

    REVIEWS {
        int id PK
        int user_id FK
        int book_id FK
        int rating
        string title
        text content
    }
}

```

2. Context-Level DFD (Level 0)

Code Specification / Diagram Snippet

```

flowchart
    User([General Reader])
    Admin([Administrator])
    GoogleBooks([Google Books API])
    System[0.0 BookSphere Platform System]

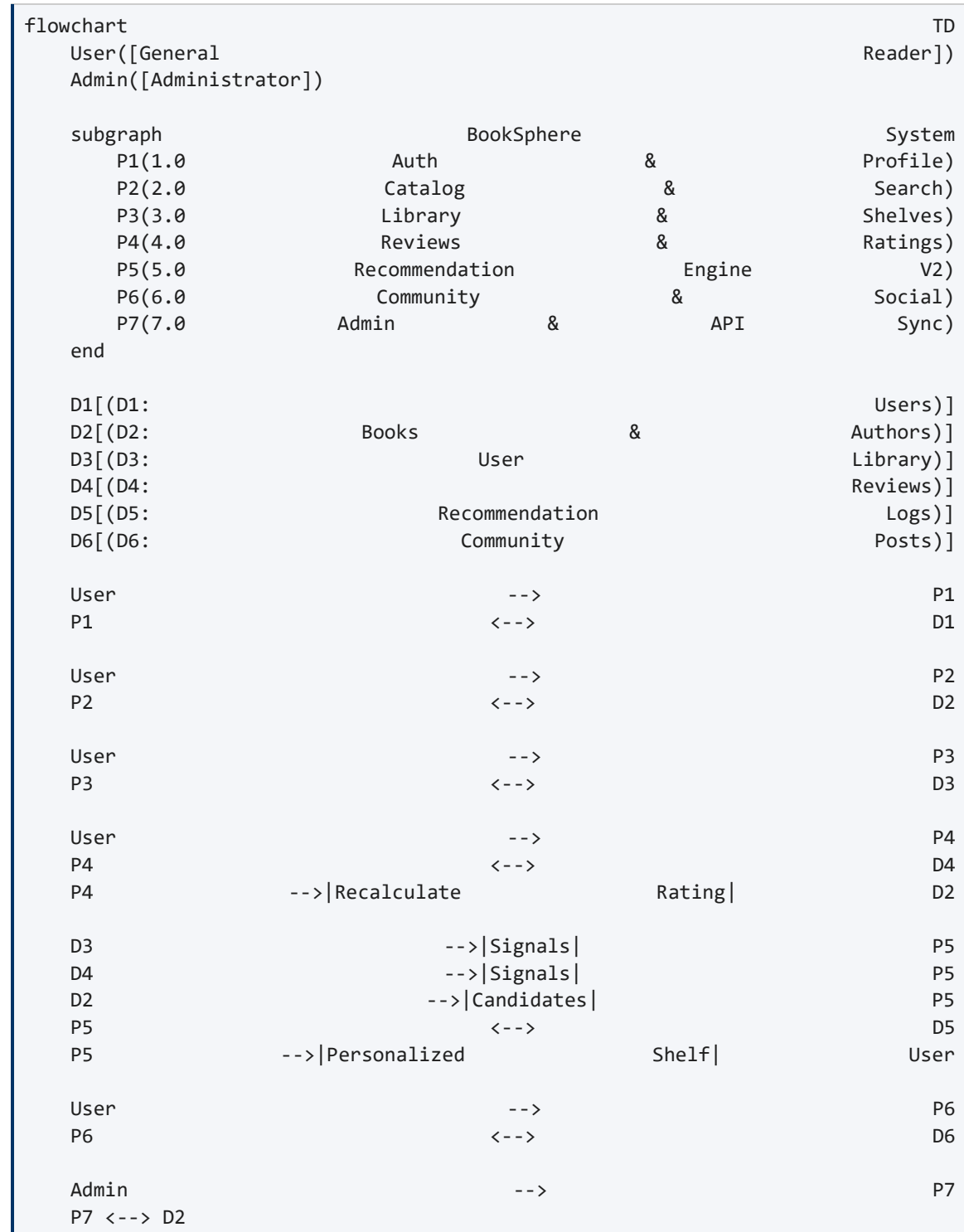
    User -->|Registration, Login, Search, Shelves, Reviews, Community| System
    System -->|Search Results, Personal Shelves, Recommendations, Feeds| User

    Admin -->|Curation, Moderation, Sync Trigger| System
    System -->|System KPIs, Reports, Moderation Queue| Admin
    
```

System	--> Volume	Search	Query,	ISBN	Lookup	GoogleBooks
GoogleBooks	--> Volume Metadata,	Cover Image	URLs	System		

3. Level-1 Data Flow Diagram (DFD)

Code Specification / Diagram Snippet



4. Use Case Diagram

Code Specification / Diagram Snippet

```

flowchart
    LR
    Guest([Guest
    Reader([Registered
    Admin([Administrator])

    subgraph BookSphere
        UC1(Browse
        UC2(Search
        UC3(Register
        UC4(Manage
        UC5(Submit
        UC6(View
        UC7(Participate
        UC8(Follow
        UC9(Admin
        UC10(Review
        UC11(Google
    end

    subgraph UseCases
        UC1(Browse
        UC2(Search
        UC3(Register
        UC4(Manage
        UC5(Submit
        UC6(View
        UC7(Participate
        UC8(Follow
        UC9(Admin
        UC10(Review
        UC11(Google
    end

    subgraph Relationships
        UC1(Browse
        UC2(Search
        UC3(Register
        UC4(Manage
        UC5(Submit
        UC6(View
        UC7(Participate
        UC8(Follow
        UC9(Admin
        UC10(Review
        UC11(Google
    end

    Guest --> UC1
    Guest --> UC2
    Guest --> UC3

    Reader -. -> |Extends| UC4
    Reader --> UC5
    Reader --> UC6
    Reader --> UC7
    Reader --> UC8

    Admin -. -> |Extends| UC9
    Admin --> UC10
    Admin --> UC11
  
```

5. Sequence Diagram: Recommendation Retrieval

Code Specification / Diagram Snippet

```

sequenceDiagram
    autonumber
    actor User
    participant Controller as RecommendationController
    participant Service as RecommendationService
    participant Repo as RecommendationRepository
    participant DB as SQLite Database

    User->>Controller: GET /recommendations
    Controller->>Service: getPersonalizedRecommendations(userId)
  
```

```

Service->>Repo:                                     userPreferences(userId)
Repo->>DB:      SELECT      library,      reviews,      wishlist
DB-->>Repo:                                     Profile      Signals
Repo-->>Service:      User      Profile      Data

Service->>Repo:                                     hybridCandidates(profileSignals)
Repo->>DB:      Parameterized      SQL      (Sources      A-H)
DB-->>Repo:      Raw      Candidate      Books
Repo-->>Service:      Candidate      Array

Service->>Service:                                     applyNegativePenalty()
Service->>Service:                                     excludeOwnLibrary()
Service->>Service:      diversityRerank(candidates,      MMR)
Service->>Service:                                     generateExplanations()
Service-->>Controller:      RecommendationResult      DTO
Controller-->>User: 200 OK (Rendered Recommendations View)

```

SCREENSHOT CHECKLIST & SPECIFICATIONS

This document provides the itemized checklist and technical specifications for the **20 verified screenshots** required for Appendix B of the academic project report.

